PurposePay: Architecture & Process Design Document

Oct 11, 2026 · @subbu

PurposePay lets a sender abroad remit money to India as digital rupee (e₹) that is locked to a purpose, such as healthcare or education, and can be spent only at merchants whose category code matches. It settles in seconds over ISO 20022 and keeps regulatory checks private with zero-knowledge proofs.

1. Executive summary

Today the retail e₹ pilot runs only inside India, and cross-border remittances still travel over SWIFT as plain cash that the sender cannot direct. PurposePay closes that gap with a middleware layer that turns an international remittance into programmable, purpose-bound e₹.

- **Sender control.** The sender splits a transfer into purpose buckets (for example 40% Education, 40% Healthcare, 20% General). Each bucket is spendable only at merchants whose 4-digit Merchant Category Code (MCC) matches.
- **Speed and cost.** Atomic settlement on a two-tier ledger targets under 3 seconds and near-zero intermediary fees, against 1–3 business days and 3–7% total cost on the SWIFT route.
- **Regulatory fit.** Messages use ISO 20022 so legacy core banking systems need no replacement, and cap checks run as zero-knowledge proofs so the ledger never exposes a sender's balance or history.

The hackathon build is a working proof of concept of the smart-contract logic and user flow. This document also sets out the production architecture, so reviewers can see how the same design scales.

2. Problem and opportunity

A remittance of USD 1,000 for medical bills shows the difference. Figures for today's route are typical ranges, not quotes from one provider.

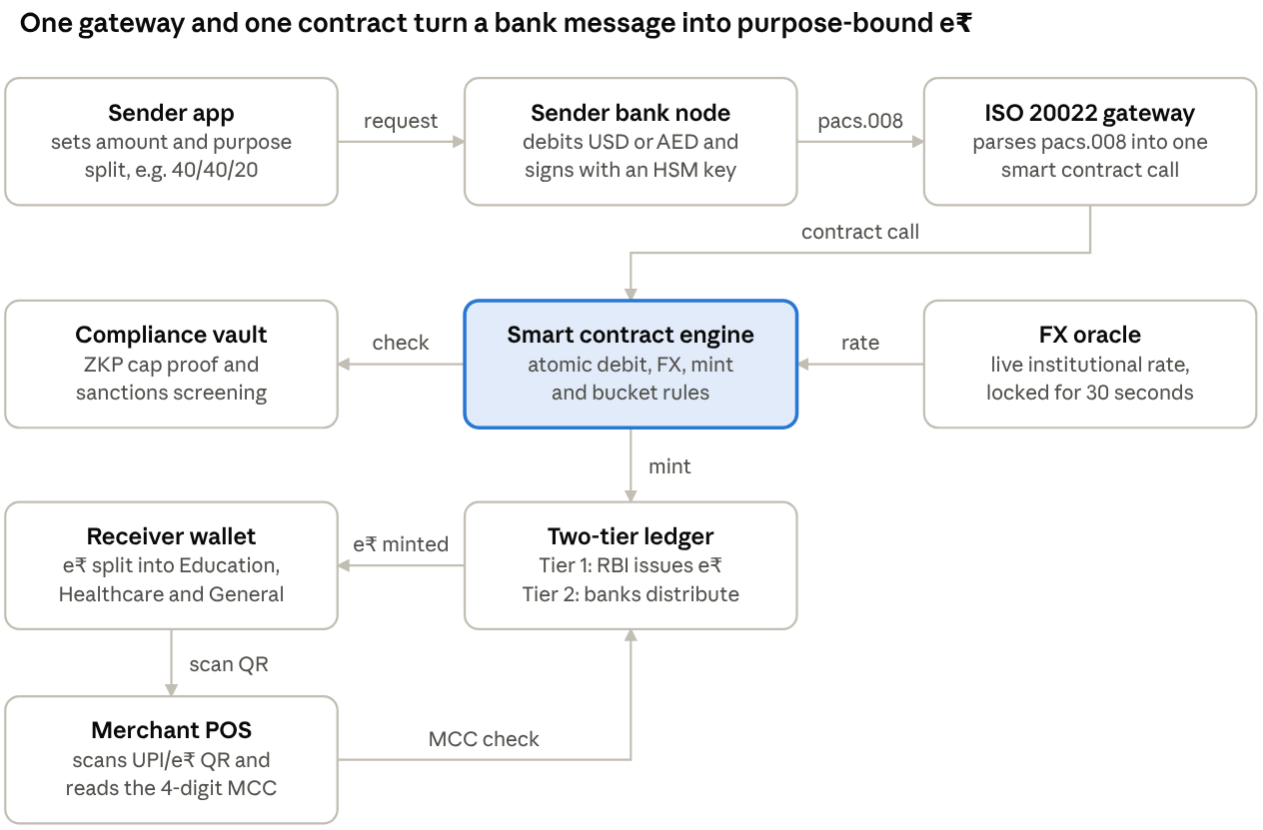
Dimension	Today (SWIFT wire)	PurposePay
Settlement time	1–3 business days across 2–4 correspondent banks	Under 3 seconds, atomic ledger settlement (target)

Dimension	Today (SWIFT wire)	PurposePay
Total cost	3–7% including wire fees and FX spread	Near-zero intermediary fees; FX at institutional rate
Messaging	Legacy MT messages	Structured ISO 20022 (pacs.008, pacs.002)
Sender control	None; funds arrive as liquid cash	Purpose buckets locked by MCC
Diversion risk	High; money can be spent anywhere	Blocked on-chain at the point of sale
Limit checks	Manual back-office review	Automated zero-knowledge proof
Visibility for sender	Ends at deposit	Spend history per bucket

Why now. Central banks, including the RBI and the Bank for International Settlements, are testing cross-border CBDC and programmable money, but only between institutions. No consumer-facing cross-border product lets a sender restrict how remitted money is spent. PurposePay demonstrates that missing piece.

3. System architecture

PurposePay is a two-tier hybrid system. The central bank issues e₹ (tier 1), commercial banks distribute it to wallets (tier 2), and one gateway and one contract tie ISO 20022 messaging to purpose rules.



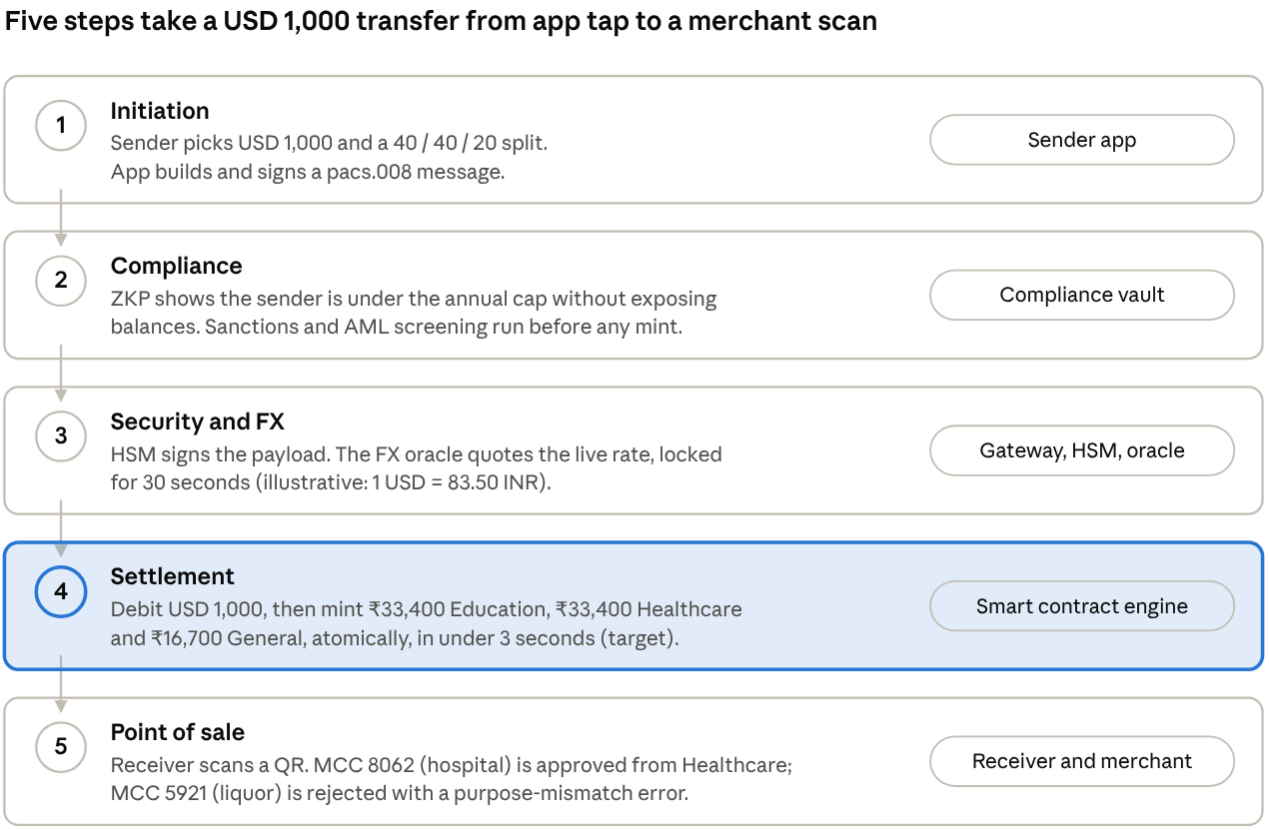
system architecture · 9 components

Read it top to bottom: the bank message enters on the first row, the contract engine checks compliance and FX, the ledger mints e₹ into the receiver's wallet, and the merchant's category code is checked back against the contract at payment.

Who plays which role. In the sender country, a commercial bank (for example Emirates NBD) debits the foreign currency and its central bank node (for example CBUAE) validates cross-currency settlement. In India, the RBI mints the purpose-bound e₹, and commercial banks (SBI, HDFC and others) host the receiver wallets, complete KYC and route merchant payments through NPCI.

4. Transaction lifecycle

The worked example is an expat in Dubai sending USD 1,000 home for medical bills, split 40% Education, 40% Healthcare and 20% General. Amounts and the FX rate are illustrative.



transaction lifecycle · 5 steps

The settlement step is the one that changes the economics. Over SWIFT, the same USD 1,000 passes through 2 to 4 correspondent banks and lands as plain cash after 1 to 3 business days. Here it lands as three locked buckets: ₹33,400 + ₹33,400 + ₹16,700 = ₹83,500 at the illustrative rate. When the contract finishes, a `pacs.002` status of `ACCC` returns to the sender's app.

5. ISO 20022 message design

The app packs the purpose split into the Supplementary Data block of a standard credit transfer, so the instruction travels through normal banking rails and is parsed by the receiving gateway without human intervention. The amounts below are illustrative.

Packs the transfer and the purpose split into `pacs.008`:

```
<Document xmlns="urn:iso:std:iso:20022:tech:xsd:pacs.008.001.10">
  <CdtTrfTxInf>
    <PmtId><EndToEndId>PURPOSEPAY-2026-98765</EndToEndId></PmtId>
    <IntrBkSttlmAmt Ccy="USD">1000.00</IntrBkSttlmAmt>
    <Dbtr><Nm>NRI Sender (Dubai)</Nm></Dbtr>
    <Cdtr><Nm>Receiver Family (India)</Nm></Cdtr>
    <SplmtryData>
      <Envlp>
        <PurposeAllocation>
          <EducationPct>40.0</EducationPct>
          <HealthcarePct>40.0</HealthcarePct>
          <GeneralPct>20.0</GeneralPct>
        </PurposeAllocation>
      </Envlp>
    </SplmtryData>
  </CdtTrfTxInf>
</Document>
```

1. **Sign.** The sender-side node signs the payload with a key held in a hardware security module (HSM), so receiving banks can detect tampering.
2. **Ingest.** The receiving gateway parses three layers: amount and currency (financial), sender and receiver identities for sanctions and AML screening (compliance), and the split percentages (rules).
3. **Invoke.** The gateway converts the message into one smart-contract call, `process_cross_border_transfer("USD", 1000.0, 40.0, 40.0, 20.0)`.
4. **Acknowledge.** After minting, the system returns a `pacs.002` status report. Status `ACCC` (accepted, settlement completed) updates the sender's app to Transfer Complete and notifies the receiver.

The custom envelope is the only PurposePay-specific part. Everything around it is standard, which is the argument for adoption: banks keep their existing core software.

6. Purpose-bound tokens and MCC enforcement

Each bucket holds e₹ tokens tagged with a purpose. When the receiver scans a merchant's UPI/e₹ QR code, the payload carries the merchant's 4-digit MCC. The smart contract approves the payment only if that MCC is in the allowed set for the bucket being spent. Merchants need no new hardware, because the MCC is already part of the standard QR payload.

Bucket	Allowed MCCs (examples)	Example blocked at checkout
Healthcare	8062 hospitals, 8011 doctors, 8021 dentists, 8099 medical services, 5912 pharmacies	5921 liquor stores, 5732 electronics
Education	8211 schools, 8220 colleges and universities, 8299 other schools and educational services	5921 liquor stores, 5732 electronics
General	Any MCC not on the regulator's prohibited list	Prohibited categories only

The lists above are a starting configuration. Production lists would be set by policy and updatable without redeploying the contract.

Valid purchase. A receiver scans an Apollo Hospital QR code (MCC 8062). The contract maps it to the Healthcare bucket and approves.

Blocked purchase. The same receiver scans a liquor store QR code (MCC 5921) with Healthcare tokens. The contract halts with the message "Transaction Rejection: Purpose-Bound Token Mismatch".

Terminal-level spoofing. A hospital cafeteria could share the hospital's MCC. In production the contract checks the parent MCC and the registered terminal or sub-merchant ID from itemised (level 3) payment data, so specialised buckets are restricted to designated billing terminals.

7. Compliance, privacy and security

Compliance runs before any token is minted, and every check is automated.

- **Limit proofs with zero-knowledge proofs (ZKP).** The sender proves that cumulative transfers plus the new amount stay under the applicable annual cap, without revealing balances or history to the shared ledger. The cap is a policy parameter. The Liberalised Remittance Scheme (LRS) limit of USD 250,000 applies to resident individuals sending money out of India; for inward NRI remittances the same proof circuit enforces the corridor and anti-money-laundering thresholds the regulator sets.
- **Sanctions and AML screening.** Sender and receiver identities are screened asynchronously against sanctions lists (such as OFAC and FIU-IND data) before execution. A hit stops the transfer before minting.
- **Identity and KYC.** The sending bank and the receiving bank each complete KYC under their own regulation. In production the receiver's wallet is linked to a national identity such as Aadhaar or a passport-backed decentralized ID.

- **Key custody.** Central-bank minting keys and bank settlement keys live in FIPS 140-2 Level 3 HSMs (for example AWS CloudHSM or Azure Dedicated HSM). Keys never leave the module.
- **Network confidentiality.** The ledger is a permissioned enterprise network (Hyperledger Fabric or Corda). Only authorised banks and central banks run nodes, and channels limit who sees each transaction.
- **Wallet recovery.** Lost phones are handled through social recovery or multi-signature guarded wallets tied to national ID, so a receiver can recover purpose-bound balances at an authorised bank branch.

8. Production readiness

The PoC proves the logic. Each PoC shortcut has a named production replacement, shown below.

Component	Hackathon PoC	Production target
Ledger and network	In-memory Python engine (`chaincode.py`)	Hyperledger Fabric or Corda Enterprise, permissioned nodes run by central and commercial banks
Key storage	SHA-256 stand-in for HSM signing	FIPS 140-2 Level 3 cloud HSMs
FX data	Static rate table	Live institutional feeds (Bloomberg B-PIPE or Refinitiv) with redundant fallback, rate locked for a 30-second window
Merchant integration	Simulated merchant registry with MCC and terminal class, enforced in the contract	Dynamic parsing of NPCI UPI/e₹ QR payloads and 4-digit MCCs
Compliance and caps	Simulated cap proof (hash commitment) against USD 250,000	On-chain ZKP verification
Throughput	Single-threaded, synchronous	Microservices on Kubernetes with an Apache Kafka event bus, targeting more than 10,000 TPS and sub-second finality with BFT consensus
Read performance	Direct state reads	Redis cache for balances and merchant permissions, written back to the ledger

Component	Hackathon PoC	Production target
Identity and KYC	Typed strings	Aadhaar or passport-linked decentralized ID

Throughput and latency numbers are design targets for production, not measured results of the PoC.

Adoption path. PurposePay is built as modular middleware, so it can run first inside existing sandboxes such as the RBI Regulatory Sandbox. Banks can issue purpose-bound synthetic tokens backed 1:1 by commercial bank money, then swap the settlement rail to central-bank e₹ once retail corridors go live. The PoC's guided demo walks this flow: a transfer, an approved hospital payment, then blocked liquor-store, cafeteria-terminal and prohibited-category payments.

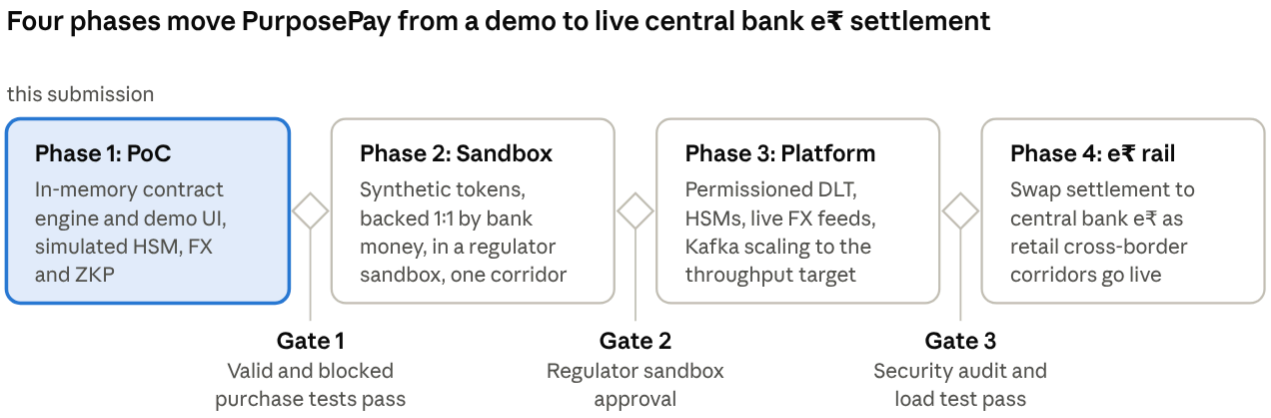
9. Risks and tough judge questions

Question or risk	Answer
"This works in a PoC. How does it work in production?"	Section 8 maps every PoC shortcut to a production component: permissioned DLT, HSMs, live FX oracles, Kafka scaling, ZKP verification.
"Central banks are slow. Why wait for them?"	The gateway is middleware. It runs today in regulatory sandboxes with purpose-bound synthetic tokens backed 1:1 by bank money, then swaps to e₹ when corridors open.
"What stops a merchant misusing its category code?"	The contract checks the parent MCC plus the registered terminal ID from itemised payment data. Acquirers already assign and audit MCCs.
"How is privacy kept while enforcing limits?"	ZKP range proofs show the cap is not exceeded without publishing balances or history.
"What if a user loses their phone?"	Social recovery or multi-signature wallets tied to national ID, with recovery at an authorised bank branch.

Question or risk	Answer
"What if a receiver needs flexibility for an emergency?"	The sender can set a General bucket share, and can reallocate unspent purpose balances through an auditable request, subject to policy.
"What if FX moves during settlement?"	The rate is locked for a 30-second execution window from the oracle feed.
"Is this legal in each corridor?"	Each corridor needs regulator approval and bilateral agreements. The policy engine keeps caps and prohibited MCC lists configurable per corridor.

10. Roadmap

The phases below are a proposed plan. Dates depend on regulator timelines, so each gate, not a calendar date, defines when the next phase starts.



roadmap · 4 phases, 3 gates

The example corridor throughout this document is UAE to India, with an NRI or expat sender as the target user.